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**ON THE LIFE-HISTORY OF *CHARACIUM*
TERRESTRIS SP. NOV.**

BY S. KANTHAMMA, B.Sc. (Hons.), M.Sc.

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ON THE LIFE-HISTORY OF *CHARACIUM TERRESTRIS* SP. NOV.*

BY S. KANTHAMMA, B.Sc. (HONS.), M.Sc.

(Communicated by M. O. P. Iyengar)

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THE alga was found growing along with *Phormidium* sp. and *Oscillatoria* sp. in some cultures of soil algae at Madras. It is unicellular ovoid or ellipsoidal and grows attached to the substratum by a more or less refractive thread-like stalk, the lower end of which is expanded into a small disc at the point of attachment. The cell contents are dense and green and it is not easy to make out the nature of the chloroplast (Figs. 1, 21-23). A single pyrenoid is imbedded in the chloroplast. Six to seven nuclei are seen around the pyrenoid in stained preparations (Figs. 2-3). Even young plants are coenocytic and contain 2-3 nuclei. The mature cell without the stalk is $26.2-38.5 \mu$ long and $22.8-38.5 \mu$ broad; the stalk is $7-10 \mu$ long.

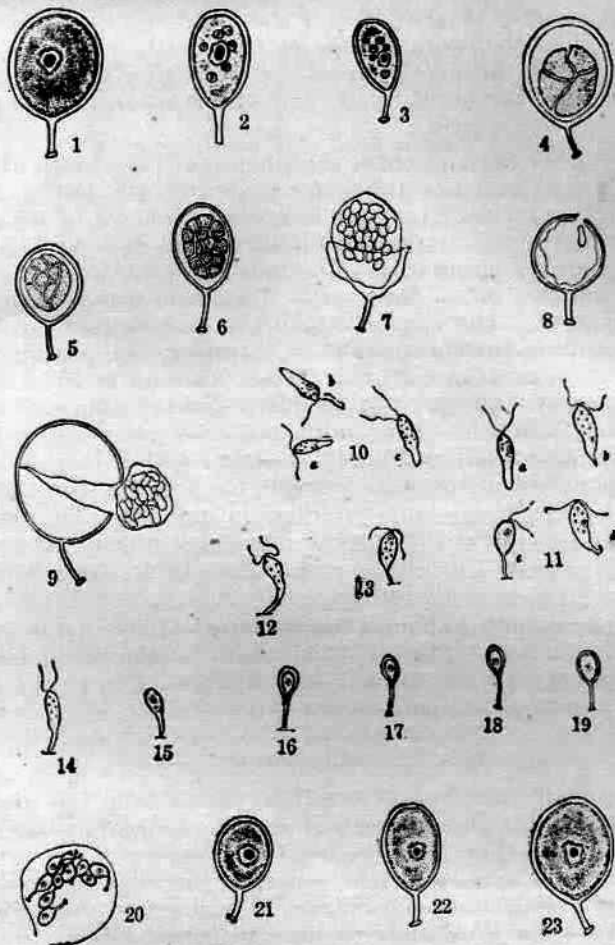
The cell-wall consists of two layers, an inner thin and delicate layer immediately next to the protoplast and an outer thicker and firm layer. The two layered nature is not very clear in the ordinary cell, but is seen very clearly in empty cells after the zoospores have escaped. Both these two layers show a cellulose reaction with iodine and sulphuric acid. These two layers are also stained deeply with ruthenium red showing that they contain plenty of pectic material as well. The stalk on the other hand shows only the cellulose reaction with iodine and sulphuric acid and does not get stained with ruthenium red showing that it is made up mainly of cellulose, without much of pectic material.

Asexual reproduction takes place by means of zoospores. Each cell by the division of its contents gives rise to about 32-64 zoospores. The first indication of the formation of the zoospores is several cleavages of the cytoplasm. The zoospores are generally liberated in the morning between 8-30 and 10 A.M. Occasionally they are liberated during mid-day also. They are biciliate and more or less conical in shape with a broadly rounded anterior end and a gradually narrowed posterior end (Figs. 10 and 11). The posterior portion is slightly hyaline and is often produced into a small tail. A very tiny papilla is present at the anterior end. The contents of the zoospore are highly granular. The chloroplast is not definite but diffuse. No pyrenoid is present. A round eye-spot is seen

* From the University Botany Laboratory, Madras.

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at the lateral region. The contractile vacuoles could not be made out clearly. A small round vacuole containing a granule is present in the anterior portion but the vacuole is not contractile (Fig. 10 b). The zoospores are $11.2-16.0\mu$ long and $3.2-4.8\mu$ broad; the cilia are about 8μ long.



Text-figs. 1-23. *Characium terrestris* sp. nov. Fig. 1. Fully grown cell. Figs. 2 and 3. Cells stained in hæmatoxylin showing the nuclei. Figs. 4-6. Cells undergoing cleavage before the formation of zoospores. Figs. 7 and 9. Zoospores escaping from the cell enclosed in a vesicle. Fig. 8. An empty cell after the escape of the zoospores. Fig. 10. a, b, c. Zoospores. Figs. 11-14. Zoospore just coming to rest; cilia still present. Figs. 15-19. Stages in the germination of the zoospores. Fig. 20. Autospores inside the vesicle. Figs. 21-23. Fully grown sporelings. All Figs. $\times 440$.

The zoospore escapes out of the mother-cell through the rupture of the cell wall across the middle or at the top (Figs. 8, 9). The cell

never gets ruptured at the stalk end. The zoospore before escaping changes its shape and position inside the mother-cell very often; it is sometimes round, sometimes elongate and sometimes contorted and finally assumes its normal shape when it escapes out. It escapes generally with the broad ciliary end foremost, but occasionally it escapes by the tail end also. Zoospores either come out one after the other or in groups of two or three. Sometimes the whole mass of zoospores covered with a thin delicate vesicle is extruded out of the mother-cell, and the zoospores then escape one by one from the vesicle.

As soon as the zoospores are liberated, they swim away from the mother-cell and are active for a short time, but during that period they swim very fast and it is very difficult to follow them. They swim with the ciliary end foremost, but the thin narrow tail end also shows a peculiar wagging movement. After swimming for about 20 minutes or so, they settle down and attach themselves to the substratum. The way in which the zoospore settles down and attaches itself to the substratum is extremely interesting and was followed in hang-drop cultures. After swimming for a time, the zoospore becomes quiescent at one place, but the cilia can be clearly seen actively vibratile. The hyaline posterior portion gets gradually still further elongated and finally becomes a thin thread-like stalk. The extreme end of the stalk attaches itself to the substratum and becomes somewhat disc shaped (Figs. 11 and 14). The cilia can be seen slowly moving at the anterior end until the stalk is completed, and then finally disappear. Whether they are discarded or absorbed inside, it could not be made out.

A young germling about a few minutes old is $6-9\ \mu$ long without the stalk and $5.2-8\ \mu$ broad. The stalk is about $8\ \mu$ long. The germling becomes a full grown plant in about a fortnight and then produces a crop of zoospores in its turn.

AUTOSPORES

Occasionally the zoospores fail to escape from the mother-cell, and develop inside the mother-wall like the ordinary germlings with a normal stalk (Fig. 20). Sometimes the zoospores which are extruded out in a mass fail to separate themselves, and develop inside the vesicle for sometime. They become free only after growing for some time and reaching a certain size.

DISCUSSION

This alga, in being unicellular and attached to the substratum by its stalk-like lower end belongs to the genus *Characium*. Its mode of asexual reproduction also shows that it is a *Characium*. It shows, however, many interesting and peculiar features which have not been so far observed in any of the previously recorded species. The mode of attachment of the zoospore to the substratum does not appear to have been observed clearly before. Smith (1916, pp. 464-65) states with regard to the zoospores of *Characium Sieboldii*, A. Br.

"Since stages in the liberation of the young zoospores were not observed the manner of their coming to rest could not be determined. From our knowledge of other algæ it would seem probable that they always come to rest with the cilia downward". The mode of attachment of the zoospores of the present alga by their narrow hyaline posterior end is extremely peculiar and quite unlike most other algæ. Coming to the mode of formation of the stalk of the alga, this appears to be very similar to that of *Ophiocytium majus* Naeg. as observed by Schiller (Pascher, 1925, p. 73). According to him the anterior part of the zoospores of *Ophiocytium majus*, during germination, becomes rounded while from the posterior portion is formed a rod-like process filled with a refractive substance. This rod-like process becomes later on the stalk of the alga. The similarity in the formation of the stalk in these two such unrelated genera, one belonging to the Chlorophyceæ and the other to the Xanthophyceæ, is a very interesting instance of parallel development.

In the mode of attachment of the zoospore to the substratum by its posterior end, the present alga resembles *Chlorangiopsis anomala* Korshikov (1932, pp. 581-82, Taf. X, Figs. 40-44), but these two algæ differ from each other in all other respects, since *Chlorangiopsis* is more nearly allied to *Chlamydomonas* than to *Characium*.

The alga appears to be a new species and may be called *Characium terrestris* sp. nov.

Characium terrestris sp. nov.

Cells shortly stalked, obovate to nearly globose, $22.8-38.5\ \mu$ broad, $26.2-38.5\ \mu$ long without stalk; stalk narrow and filamentous $7-10.5\ \mu$ long and expanded at the point of attachment into a tiny disc. Zoospores biciliate and conical with a broadly rounded anterior end and a narrow elongate posterior end, $3.2-4.8\ \mu$ at the broadest part and $11.2-16\ \mu$ long.

Hab.—In a laboratory culture of soil algæ at Madras.

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